

trimly

Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for trimly, a dual-sided platform that enables customers to discover salons, preview styles using AI, book appointments, make payments, earn rewards, and review services, while enabling salon managers to manage operations including scheduling, staff, inventory, suppliers, and analytics.

1.2 Scope

The platform trimly consists of:

- **Consumer Mobile App (React Native + Expo):** Salon discovery, booking, payments (PayHere), AI preview, reminders, loyalty rewards, reviews, profile and appointment history.
- **Business Web Portal (Next.js):** Salon management (services, stylists, schedules, staff shifts, clients), inventory, dashboard, revenue insights.
- **Backend API (Node.js/NestJS):** Authentication, data isolation across salons, booking logic, payments, notifications, AI integration and analytics.

1.3 Definitions and Acronyms

- **SRS:** Software Requirements Specification
- **RBAC:** Role-Based Access Control
- **Tenant / Multi-tenant:** Multiple salons share the platform while keeping data isolated
- **MVP:** Minimum Viable Product
- **AI Style Mirror:** AI feature to generate hairstyles/colors on a user photo

1.4 Assumptions and Dependencies

- Users require internet access to use the platform.
- Payments are processed through a third-party payment provider (e.g., PayHere).
- AI services are integrated via external APIs; custom model training is out of scope.
- Inventory “tracking” is based on manager updates and stock logs (no automated hardware integration).

2. Overall Description

2.1 Product Perspective

The platform trimly is a multi-tenant: many salons operate within the same system. Each salon's data is isolated by a salonId scope. Consumers can browse multiple salons; managers can manage only their own salon.

2.2 User Classes and Characteristics

- **Customer (Mobile App):** Books services, pays, receives reminders, earns loyalty points, views the history, leaves reviews, uses AI preview.
- **Salon Manager (Web Portal):** Manages the salon profile, services, stylists, staff shifts, appointments, inventory, suppliers, analytics.

2.3 Operating Environment

- **Mobile:** iOS/Android via React Native + Expo
- **Web:** Modern browsers (Chrome, Edge, Firefox)
- **Backend:** Cloud-hosted REST API
- **Database:** PostgreSQL (primary), Redis optional for caching

2.4 Constraints

- Development timeline: 4 months
- Must support multi-salon data isolation
- Must provide secure authentication and payment handling
- Must provide responsive, premium user experience

3. Functional Requirements

Priority Definitions:

- **Must:** required for core system operation
- **Should:** important but can be deferred if needed
- **Could:** optional enhancement

3.1 Authentication & Roles

FR-01 — Customer Registration/Login (Must)

The system shall allow customers to register and log in using supported credentials.

Acceptance Criteria: Customer can create an account, log in, log out, and access protected screens.

FR-02 — Salon Manager Login (Must)

The system shall allow salon managers to authenticate and access the business portal.

Acceptance Criteria: Manager can log in and access only management functions.

FR-03 — Role-Based Access Control (Must)

The system shall restrict features based on role (Customer vs Manager).

Acceptance Criteria: Customers cannot access manager portal routes; managers cannot access customer-only endpoints.

3.2 Salon Discovery & Details (Customer)

FR-04 — Browse Salons (Must)

The system shall display a list of salons to customers with basic information (name, location, rating summary if available).

Acceptance Criteria: Customer can view salon list and select a salon for details.

FR-05 — View Salon Details (Must)

The system shall allow customers to view salon services, pricing, estimated durations, and stylist list.

Acceptance Criteria: Services display price and duration; stylists are visible; salon info is readable.

FR-06 — Search/Filter Salons (Should)

The system shall allow customers to search/filter salons by location and service category.

Acceptance Criteria: Filters update results and can be cleared.

3.3 Booking & Scheduling

FR-07 — View Availability (Must)

The system shall show available time slots for a selected salon/service/date.

Acceptance Criteria: Unavailable slots are not selectable; available slots reflect current bookings.

FR-08 — Book Appointment (Must)

The system shall allow customers to book an appointment by selecting service, date/time, and optional stylist.

Acceptance Criteria: Booking succeeds only if slot is available; confirmation is generated.

FR-09 — Prevent Double Booking (Must)

The system shall prevent booking conflicts for the same stylist/time slot.

Acceptance Criteria: Conflicting booking attempts are rejected with an appropriate message.

FR-10 — Reschedule/Cancel Appointment (Should)

The system shall allow customers (and managers) to reschedule or cancel appointments according to salon policy.

Acceptance Criteria: Changes update availability and notify involved parties.

3.4 Payments

FR-11 — Payment Processing (Must)

The system shall allow customers to pay for appointments using a payment provider.

Acceptance Criteria: Payment success/failure is recorded; user receives confirmation.

FR-12 — Payment Status & Receipts (Must)

The system shall store payment status (paid/unpaid/refunded) per appointment and provide a receipt/summary.

Acceptance Criteria: Appointment shows correct payment status; manager can view transaction summaries.

FR-13 — Refund Handling (Should)

The system shall support refunds through the payment provider for eligible cancellations.

Acceptance Criteria: Refund is recorded and reflected in reports.

3.5 Notifications

FR-14 — Booking Confirmation (Must)

The system shall send booking confirmation notifications after successful booking/payment.

Acceptance Criteria: Customer receives confirmation with correct appointment details.

FR-15 — Appointment Reminders (Must)

The system shall send reminder notifications prior to scheduled appointments.

Acceptance Criteria: Reminder sent at configured times (e.g., 24 hours, 1 hour); includes date/time and salon.

3.6 AI Features**FR-16 — AI Style Preview (Should)**

The system shall allow customers to upload a photo and preview hairstyle/color options via AI.

Acceptance Criteria: Photo upload works; AI preview result is displayed; errors are handled gracefully.

FR-17 — Personalized Recommendations (Should)

The system shall suggest services based on customer history/preferences and salon offerings.

Acceptance Criteria: Recommendations are shown with a short reason and can lead to booking.

3.7 Reviews & Ratings**FR-18 — Submit Review (Must)**

The system shall allow customers to leave a rating and written review for a completed appointment.

Acceptance Criteria: Reviews can only be submitted for completed/eligible appointments; rating stored 1–5.

FR-19 — View Reviews (Must)

The system shall display reviews and an overall rating summary on the salon profile.

Acceptance Criteria: Reviews list visible; average rating updates based on stored reviews.

FR-20 — Review Moderation (Could)

The system may allow managers to report reviews for moderation.

Acceptance Criteria: Report reason captured; flagged review tracked.

3.8 Business Portal Requirements (Salon Manager)

FR-21 — Manage Salon Profile (Must)

The system shall allow managers to create/update salon information (name, address, hours, images).

Acceptance Criteria: Changes persist and are visible on customer side.

FR-22 — Manage Services (Must)

The system shall allow managers to add/edit/remove services including price and duration.

Acceptance Criteria: Updated services appear in booking flow.

FR-23 — Manage Stylists (Must)

The system shall allow managers to create and manage stylist profiles and assign them to services.

Acceptance Criteria: Stylists can be activated/deactivated; bookings reflect current stylist availability.

FR-24 — Manage Calendar (Must)

The system shall provide a calendar view for appointments and allow rescheduling/cancellation.

Acceptance Criteria: Calendar shows appointments; changes update availability.

FR-25 — Client Profiles & Notes (Should)

The system shall allow managers to store client history and notes.

Acceptance Criteria: Notes are saved per client and visible to managers.

3.9 Inventory & Suppliers

FR-26 — Manage Inventory Items (Must)

The system shall allow managers to create inventory items with stock quantity, unit, and reorder threshold.

Acceptance Criteria: Items can be created/edited; current quantity displayed.

FR-27 — Record Stock Adjustments (Must)

The system shall allow managers to manually update stock when items are used or received. Examples: “Used 1 bottle”, “Received 20 units”.

Acceptance Criteria: Each adjustment updates quantity and creates a stock log entry.

FR-28 — Inventory Usage Log (Should)

The system shall maintain an audit trail of inventory adjustments (date, reason, quantity, user).

Acceptance Criteria: Logs are viewable and filterable.

FR-29 — Low Stock Alerts (Must)

The system shall generate alerts when stock falls below a reorder threshold.

Acceptance Criteria: Alert appears in dashboard and/or notifications.

FR-30 — Manage Suppliers (Must)

The system shall allow managers to add and maintain supplier records for the salon.

Acceptance Criteria: Supplier name/contact/details stored and editable.

FR-31 — Link Inventory Items to Suppliers (Should)

The system shall allow inventory items to be associated with a preferred supplier.

Acceptance Criteria: Each item can reference supplier; supplier can be changed.

FR-32 — Create Inventory Orders (Could)

The system may allow managers to create simple purchase orders to suppliers for restocking.

Acceptance Criteria: Order captures items/quantities and status (draft/sent/received).

3.10 Analytics & Reports

FR-33 — Revenue Summary (Must)

The system shall provide daily and monthly revenue totals and appointment counts.

Acceptance Criteria: Totals match stored payment/appointment data.

FR-34 — Staff Performance Summary (Should)

The system shall show service counts per stylist and commission calculations if configured.

Acceptance Criteria: Metrics are calculated by the selected date range.

3.11 Loyalty & Rewards

FR-35 — Loyalty Points Accumulation (Should)

System shall allow customers to earn points for completed appointments.

Acceptance Criteria: Loyalty points are automatically added to the customer's account after an appointment is marked as completed.

FR-36 — Redeem Loyalty Rewards (Could)

System shall allow customers to redeem points for discounts or offers.

Acceptance Criteria: Customers can apply available loyalty points to eligible bookings and see the updated discount before confirming payment.

3.12 Multi-Tenancy

FR-37 — Data Isolation Between Salons (Must)

The system shall ensure managers can only access data belonging to their salon.

Acceptance Criteria: Cross-salon access attempts are denied; queries scoped by salon.

4. External Interface Requirements

4.1 Mobile App Interfaces (Expo)

Key screens:

- Login/Register
- Salon List + Filters
- Salon Details (services, stylists, reviews)
- Booking Flow
- Payments
- AI Style Preview
- Appointment History
- Profile/Settings

4.2 Web Portal Interfaces (Next.js)

Key pages:

- Manager Login
- Dashboard
- Salon Profile
- Services Management
- Stylists Management
- Calendar
- Clients/CRM
- Inventory & Suppliers
- Reports/Analytics

4.3 API Interfaces (Backend)

Representative endpoints:

- Auth: /auth/register, /auth/login
- Salons: /salons, /salons/{id}
- Services: /salons/{id}/services
- Stylists: /salons/{id}/stylists
- Appointments: /appointments
- Payments: /payments/checkout, /payments/webhook
- Reviews: /salons/{id}/reviews
- Inventory: /inventory/items, /inventory/adjustments
- Suppliers: /suppliers

4.4 Third-Party Integrations

- Payment provider (e.g., Stripe)
- AI API provider(s)
- Push notifications (Expo Notifications)
- Email/SMS provider (optional fallback)

5. Non-Functional Requirements

NFR-01 Performance (Must)

- Mobile screens should load primary content within 3 seconds on typical mobile networks.
- API responses for common endpoints should average under 500ms under normal load.

NFR-02 Availability (Should)

- The system should target high availability for production deployment (aspirational 99.9%).

NFR-03 Security (Must)

- All traffic shall use HTTPS.
- Passwords shall be hashed securely.
- JWT tokens shall be validated for protected endpoints.

- RBAC shall be enforced for all sensitive actions.

NFR-04 Privacy (Must)

- User photos used for AI preview shall be stored securely or processed transiently depending on implementation.
- Customers should be able to request deletion of their data (basic compliance support).

NFR-05 Reliability & Data Integrity (Must)

- Booking operations shall prevent conflicting appointments.
- Payment status shall be consistent with provider callbacks (webhooks).

NFR-06 Scalability (Should)

- System architecture shall support onboarding additional salons without redesign.

NFR-07 Usability (Must)

- Mobile booking flow should be achievable within a minimal number of steps.
- UI should remain responsive and accessible.

NFR-08 Maintainability (Should)

- Codebase shall use consistent linting and formatting.
- The system should follow modular structure and clear separation of concerns.

NFR-09 Compatibility (Must)

- Mobile: Android and iOS supported via Expo.
- Web: Compatible with latest versions of major browsers.

NFR-10 Scalability / Architecture Readiness (Should)

- The backend shall be designed using a modular monolith or microservice-ready structure to support future scaling.

NFR-11 UI/UX Aesthetics (Must)

- The UI shall follow an “Apple-esque” premium design with smooth animations; web animations may use Framer Motion where appropriate.

6. Data Requirements

Core entities (minimum):

- User (Customer, Manager)
- Salon
- Service
- Stylist (entity, not necessarily a login role in MVP)
- Appointment
- Payment Transaction
- Review
- InventoryItem
- InventoryAdjustmentLog
- Supplier

Multi-tenant support:

- Salon-related entities include salonId to enforce isolation.

7. Out of Scope (Current Phase)

- Custom ML model training
- Multi-branch salon management
- Automated inventory reduction via IoT/barcode scanning

8. Verification & Acceptance

The system will be validated via:

- End-to-end demo flows: browse → book → pay → reminder → review
- Manager flows: manage services/stylists → calendar → inventory adjustments → supplier setup
- API testing using Postman
- Basic unit/integration tests for booking conflict and payment status updates